IIT JAMMU SUMMER SCHOOL 2026

ASSIGNMENT

Course: IOT, Robotics and Drones

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Course: IoT, Robotics and Drones

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Q1. What is a drone (UAV)? Explain its definition, history, and evolution?

ANSWER1: 

A drone, also known as an Unmanned Aerial Vehicle (UAV), is an aircraft that operates without a human pilot on board. It can be controlled remotely by a human operator or fly autonomously using onboard computers, sensors, GPS, and artificial intelligence. UAVs are used for a wide range of applications, including surveillance, mapping, agriculture, photography, package delivery, disaster management, and military operations.

Definition

A UAV is defined as:

"An aircraft that is capable of sustained flight without a pilot onboard and is controlled either remotely or autonomously through pre-programmed flight plans and onboard systems."

A complete drone system typically includes:

- UAV (aircraft): The flying platform.
- Ground Control Station (GCS): Used by the operator to monitor and control the UAV.
- Communication Link: Transfers commands and telemetry between the UAV and the GCS.
- Payload: Equipment carried by the UAV, such as cameras, sensors, GPS receivers, or spraying systems.

History of Drones (UAVs)

The development of UAVs spans more than a century.

1. Early Concepts (1849–1913)

- 1849: Austria used unmanned balloons carrying explosives against Venice, representing one of the earliest unmanned aerial attacks.
- 1898: Nikola Tesla demonstrated radio-controlled technology, laying the foundation for remotely operated vehicles.
- 1913: The first experiments with automatic flight control systems began.

2. World War I (1914–1918)

- Development of pilotless aircraft accelerated.
- The Kettering Bug (1918), an early unmanned "aerial torpedo," was one of the first practical UAV concepts.

3. World War II (1939–1945)

- Radio-controlled target drones were developed for military training.
- Germany introduced the V-1 flying bomb, an early cruise missile that influenced future unmanned aircraft development.

4. Cold War Era (1950s–1980s)

- UAVs were primarily used for military reconnaissance and intelligence gathering.
- Improvements in electronics, cameras, and communication systems made drones more reliable.

5. Modern Military UAVs (1990s–2000s)

- GPS, satellite communication, and digital imaging transformed UAV capabilities.
- Aircraft such as the MQ-1 Predator became widely used for surveillance and precision missions.

6. Civilian Drone Revolution (2010–Present)

- Advances in batteries, lightweight materials, GPS, and smartphones made drones affordable.
- Commercial drones are now widely used in:
  - Aerial photography and filmmaking
  - Precision agriculture
  - Infrastructure inspection
  - Surveying and mapping
  - Disaster response
  - Package delivery
  - Environmental monitoring

Evolution of UAV Technology

The evolution of drones can be divided into several stages:

Generation	Characteristics
First Generation	Simple radio-controlled target aircraft with limited capabilities.
Second Generation	Military reconnaissance drones equipped with cameras and longer flight endurance.
Third Generation	GPS navigation, autonomous flight, real-time communication, and improved sensors.
Fourth Generation	AI-powered drones featuring obstacle avoidance, computer vision, swarm coordination, and autonomous decision-making.

Major Technological Advances

- Propulsion: Improved electric motors, fuel-efficient engines, and high-capacity batteries.
- Navigation: GPS, inertial navigation systems (INS), and real-time positioning.
- Sensors: High-resolution cameras, thermal imaging, LiDAR, multispectral sensors, and radar.
- Communication: Radio frequency (RF), Wi-Fi, 4G/5G, and satellite communication.
- Autonomy: Artificial intelligence, machine learning, obstacle detection, autonomous takeoff/landing, and swarm technology.

Present-Day Applications

Modern UAVs are widely used in:

- Defense and border surveillance
- Precision agriculture
- Disaster management and search-and-rescue
- Environmental monitoring
- Surveying and mapping
- Infrastructure inspection (bridges, power lines, pipelines)
- Wildlife conservation
- Logistics and package delivery
- Aerial photography and cinematography

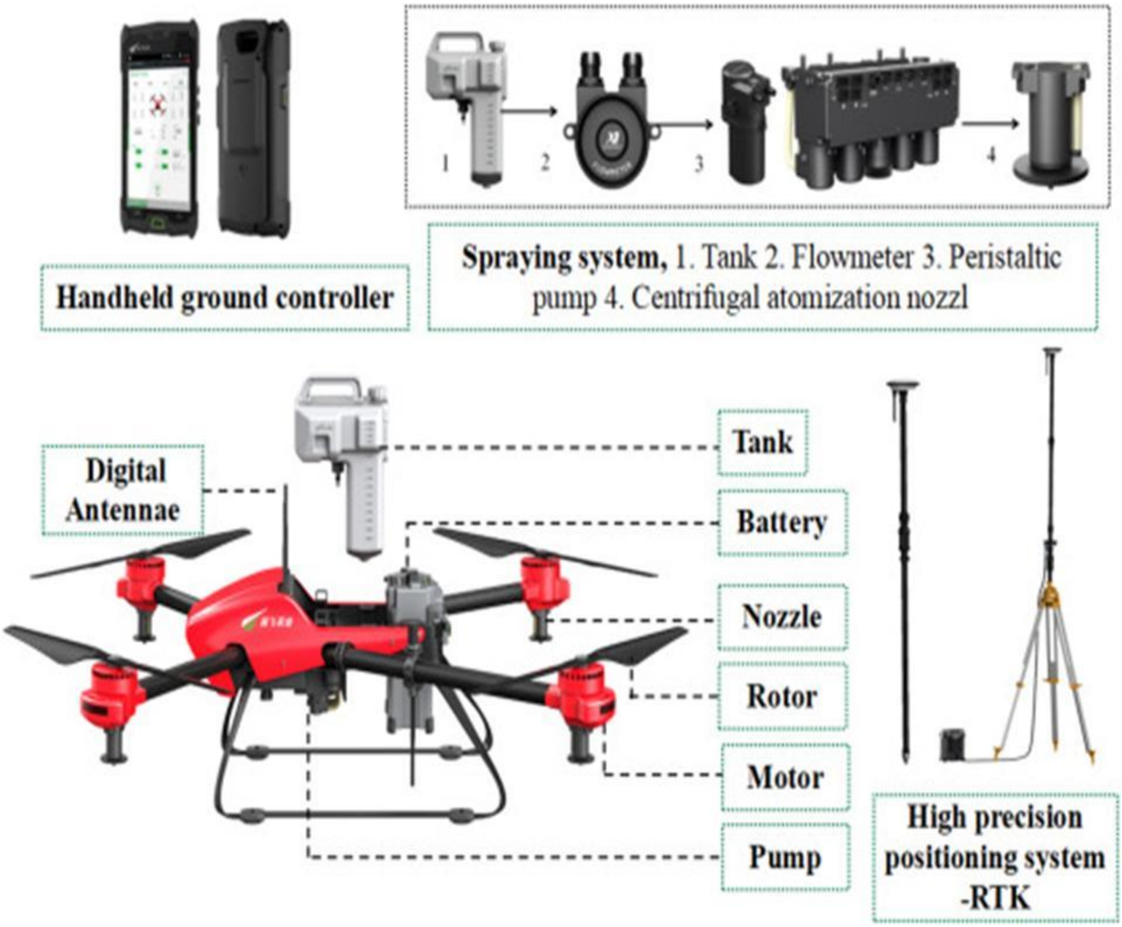
Conclusion

A drone (UAV) is an unmanned aircraft that can be operated remotely or autonomously. From early military experiments to today's AI-enabled autonomous systems, UAV technology has evolved dramatically. Modern drones combine advanced sensors, GPS, communications, and intelligent flight control to perform tasks that are often safer, faster, and more cost-effective than conventional manned aircraft. Their applications continue to expand across defense, industry, agriculture, scientific research, and everyday commercial use.

Q2: List at least five real-world applications of drones?

ANSWER2: Drone (Unmanned Aerial Vehicle, UAV):
A drone is an aircraft that operates without an onboard pilot. It is controlled remotely or flies autonomously using technologies such as GPS (Global Positioning System), IMU (Inertial Measurement Unit), flight controllers, computer vision, and wireless communication systems.

Agriculture (Precision Agriculture):



Applications

- Crop health monitoring using multispectral and thermal cameras
- Precision spraying of fertilizers and pesticides
- Irrigation monitoring
- Crop yield estimation

Technical Terminology

- NDVI (Normalized Difference Vegetation Index): Measures plant health using reflected light.

- RTK GPS (Real-Time Kinematic GPS): Provides centimeter-level positioning accuracy.
- Variable Rate Application (VRA): Applies inputs only where needed.

Real-World Example

DJI Agras agricultural drones are widely used in China, India, and many other countries for automated pesticide spraying, reducing chemical use and improving productivity.

Advantages

- Reduces labor costs
- Saves water and chemicals
- Detects diseases early
- Improves crop yield

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2. Disaster Management and Search & Rescue

Applications

- Locating missing persons
- Flood monitoring
- Forest fire detection
- Earthquake damage assessment
- Delivering emergency medical supplies

Technical Terminology

- Thermal Imaging Camera
- LiDAR (Light Detection and Ranging)
- Geotagging
- Real-Time Video Transmission

Real-World Example

National Disaster Response Force (India) has used drones during flood and landslide operations for aerial surveillance and victim identification.

Advantages

- Rapid deployment
- Improves responder safety
- Covers inaccessible areas
- Provides real-time situational awareness

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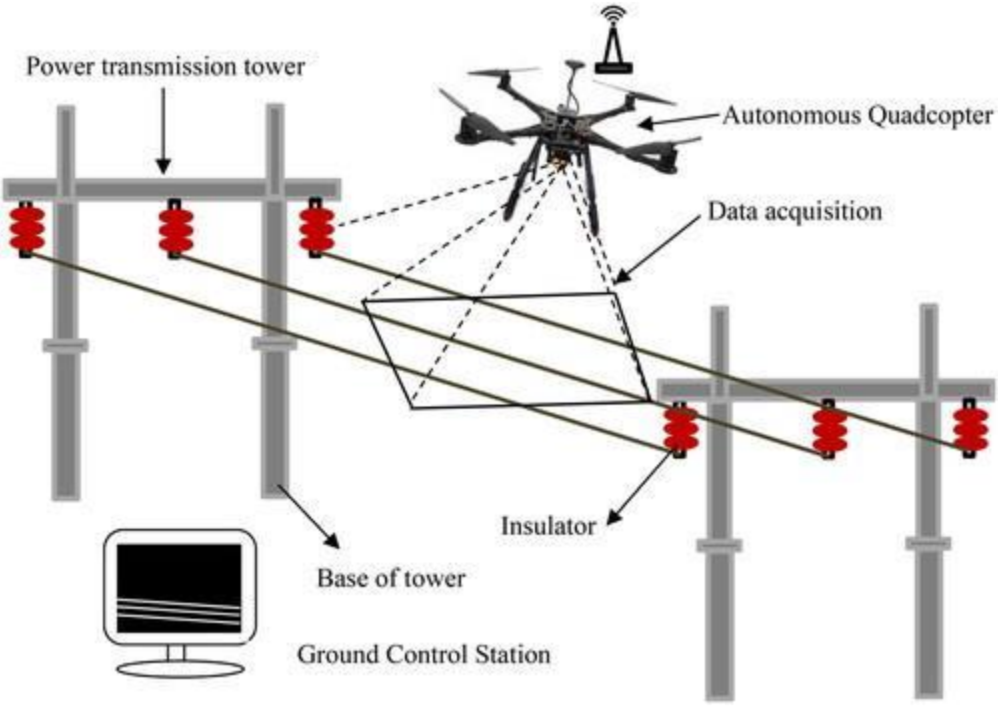
3. Infrastructure Inspection



Applications

- Inspection of bridges
- Power transmission lines
- Wind turbines
- Pipelines
- Railway tracks

Technical Terminology

- High-resolution RGB Camera
- Photogrammetry
- 3D Mapping
- Digital Twin

Real-World Example

National Grid plc uses drones to inspect high-voltage power lines, reducing the need for workers to climb towers and improving maintenance efficiency.

Advantages

- Safer inspections
- Reduced downtime
- High-quality imaging
- Lower operational costs

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4. Logistics and Medical Delivery

Applications

- Medicine delivery

- Blood sample transportation
- Vaccine delivery
- Emergency supplies
- E-commerce parcel delivery

Technical Terminology

- Autonomous Navigation
- BVLOS (Beyond Visual Line of Sight)
- Payload Capacity
- Flight Management System

Real-World Example

Zipline delivers blood, vaccines, and medicines by drone in Rwanda, Ghana, and other countries, significantly reducing delivery times to remote healthcare facilities.

Advantages

- Faster emergency deliveries
- Access to remote regions
- Reduced transportation costs
- Improved healthcare services

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5. Surveying, Mapping, and Construction

Applications

- Land surveying
- Topographic mapping
- Construction progress monitoring
- Mine surveying
- Volume measurement

Technical Terminology

- Photogrammetry
- Orthomosaic Map
- Digital Elevation Model (DEM)
- Ground Control Points (GCPs)

Real-World Example

senseFly mapping drones are used by engineering and surveying firms worldwide to create accurate 3D terrain models and construction site maps.

Advantages

- High measurement accuracy
- Faster surveys
- Reduced fieldwork
- Better project planning

Q3: *Differentiate between:(a) UAV and Drone,(b) Fixed-wing and Multirotor drones,(c)Quadcopter and Hexacopter?*

Answer3:

(a) Difference Between UAV and Drone

UAV (Unmanned Aerial Vehicle)	Drone
A UAV is an aircraft that flies without a human pilot onboard.	A drone is a broader term commonly used for any unmanned vehicle, especially flying ones.
Refers specifically to the aircraft itself.	May refer to the complete system, including the UAV, controller, sensors, and communication equipment.
Used mainly in technical, military, and aviation contexts.	Commonly used in commercial, recreational, and general contexts.
Can be remotely piloted or autonomous.	Can also be remotely piloted or autonomous.
Example: A military surveillance UAV.	Example: A DJI camera drone used for aerial photography.

DIAGRAM OF UAV

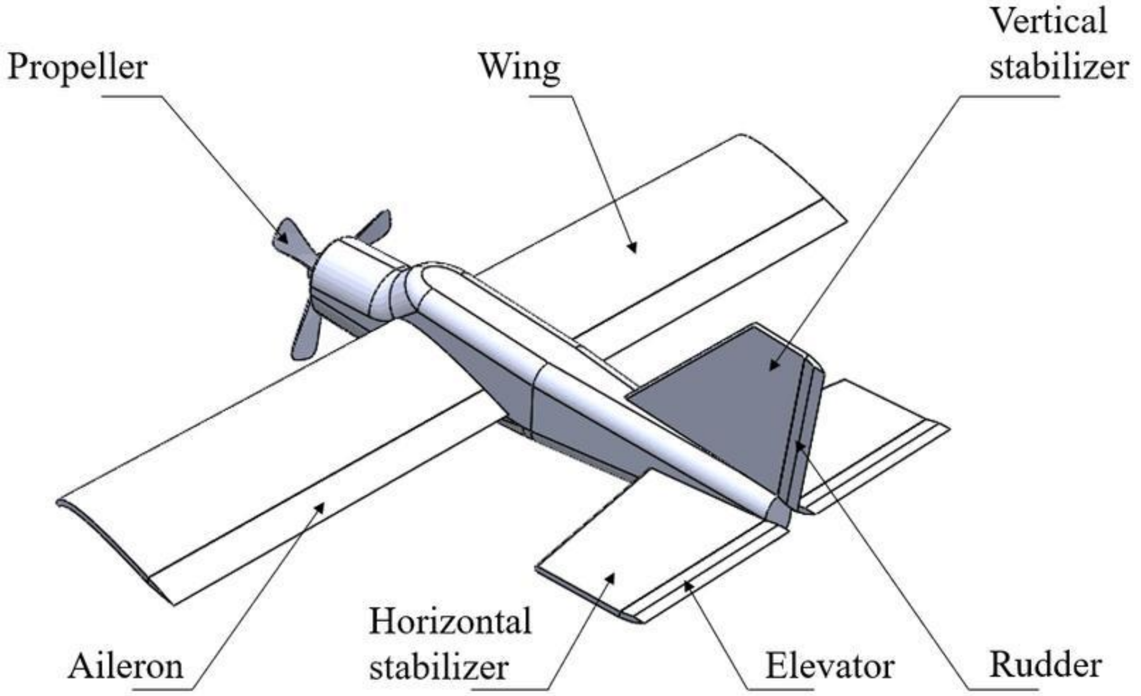
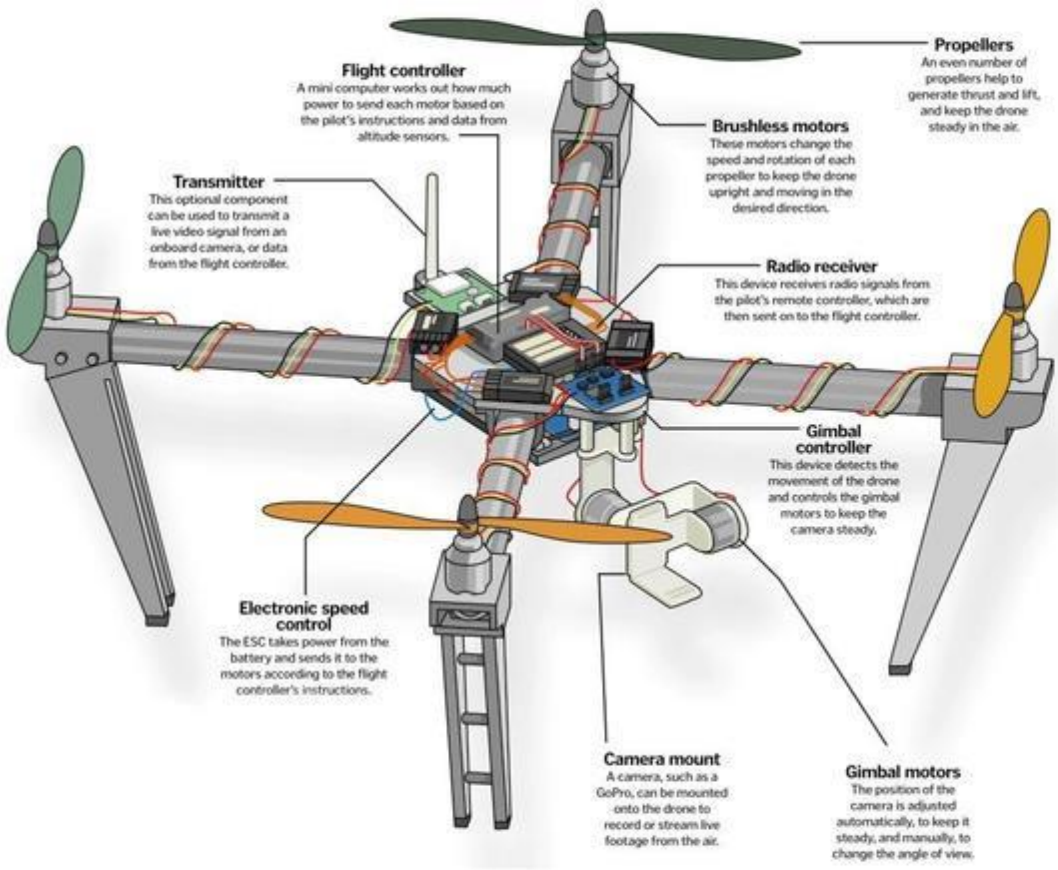


DIAGRAM OF DRONES



(b) Difference Between Fixed-Wing and Multirotor Drones

Fixed-Wing Drone	Multirotor Drone
Has airplane-like wings that generate lift.	Uses multiple rotors (propellers) to generate lift.
Cannot hover in one place.	Can hover and fly in any direction.
Longer flight time (typically 1–10 hours depending on model).	Shorter flight time (typically 20–60 minutes).
Covers large distances efficiently.	Best for close-range operations and precise maneuvering.
Requires a runway or launcher for takeoff (except VTOL models).	Takes off and lands vertically (VTOL).
Faster and more energy-efficient.	Slower but highly maneuverable.
Applications: Mapping, surveillance, agriculture.	Applications: Photography, inspection, delivery, rescue.

DIAGRAM OF FIXED WING

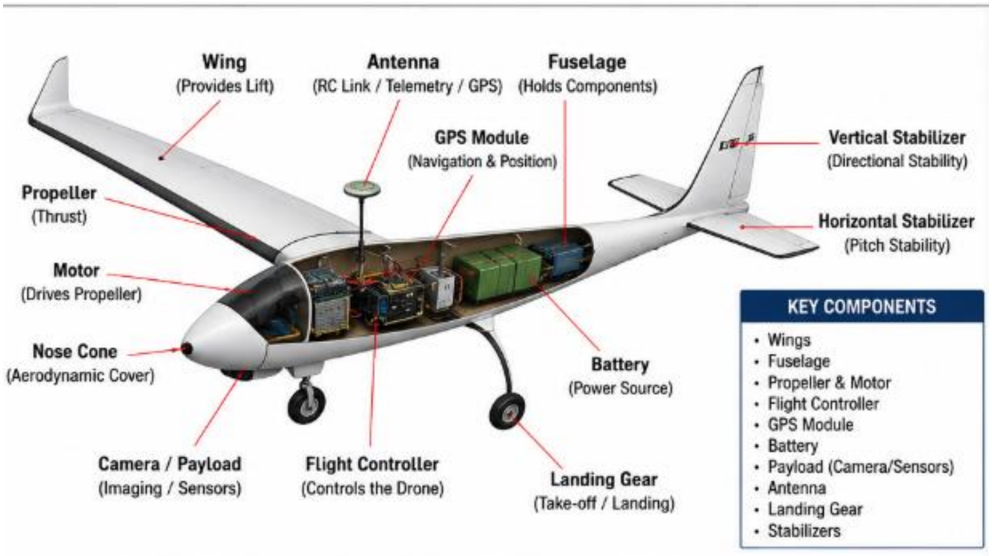
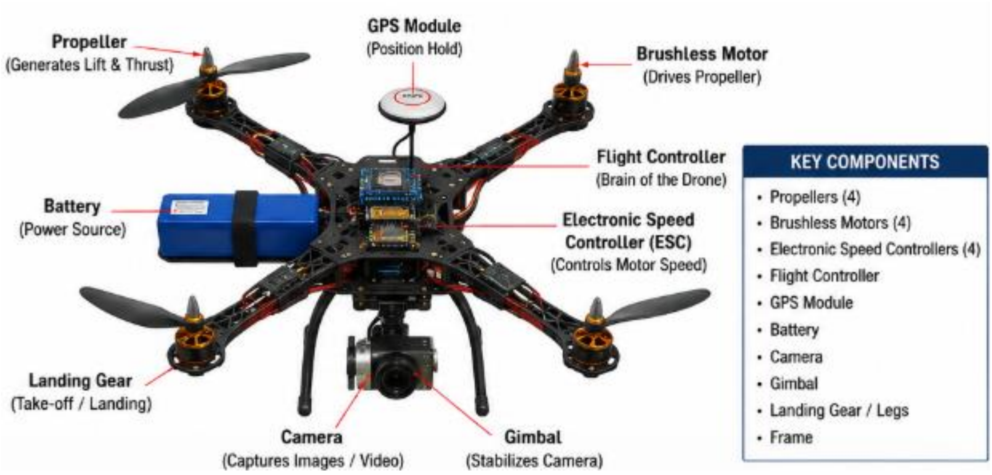


DIAGRAM OF MUTIROTOR DRONES



(c) Difference Between Quadcopter and Hexacopter

Quadcopter	Hexacopter
Has 4 motors and 4 propellers.	Has 6 motors and 6 propellers.
Simpler design and lower cost.	More complex and more expensive.
Lower payload capacity.	Higher payload capacity for heavier equipment.
Less stable in strong winds.	Greater stability and smoother flight.
Failure of one motor usually causes a crash.	Can often continue flying safely even if one motor fails.
Lower power consumption.	Higher power consumption due to extra motors.

Applications: Hobby flying, aerial photography, inspections.	Applications: Professional cinematography, LiDAR surveys, industrial inspection, heavy camera payloads

DIAGRAM OF QUADCOPTER:

DIARAM OF HEXACOPTER

Q5. Explain how a drone performs take-off, landing, hovering, move forward, move backward, move left, move right, and rotate (yaw). Use suitable diagrams?

ANSWER5: A quadcopter drone controls its movement by changing the speed (RPM) of its four brushless motors. The flight controller continuously adjusts the motor speeds to maintain stability and achieve the desired motion.

1. Take-off

During take-off, all four motors increase their speed equally. When the total upward thrust becomes greater than the drone's weight, the drone rises vertically.

2. Landing

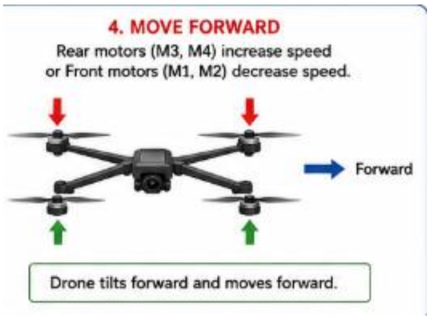
During landing, all four motors decrease their speed equally. The total thrust becomes less than the drone's weight, allowing it to descend smoothly.

3. Hovering

To hover, all four motors rotate at the same speed, producing thrust exactly equal to the drone's weight.

4. Move Forward

The rear motors (M3 and M4) rotate faster, while the front motors (M1 and M2) rotate slightly slower. This tilts the drone forward, causing forward motion.



5. Move Backward

The front motors (M1 and M2) rotate faster, while the rear motors (M3 and M4) rotate slower. The drone tilts backward and moves backward.

6. Move Left

The right motors (M2 and M4) rotate faster, while the left motors (M1 and M3) rotate slower. The drone tilts left and moves left.

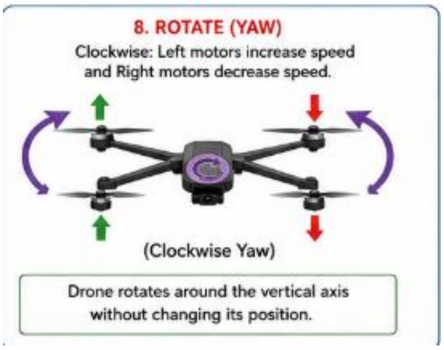
7. Move Right

The left motors (M1 and M3) rotate faster, while the right motors (M2 and M4) rotate slower. The drone tilts right and moves right.

8. Rotate (Yaw)

Yaw is the rotation of the drone around its vertical axis without changing its position.

- For clockwise yaw, the CCW motors (M1 and M4) increase speed while the CW motors (M2 and M3) decrease speed.
- For counter-clockwise yaw, the opposite occurs.



Q6. Draw a labeled block diagram of a drone and explain the function of each component: flight controller, ESC, brushless motors, propellers, battery, power distribution board, GPS module, radio receiver, frame, telemetry module?

Answer6: Labeled Block Diagram of a Drone

```
                    GPS Module
                        |
                        ▼
              +-------------------------+
              | Flight controller       |
              +-------------------------+

                  ▲                ▲
                  |                |
  Radio Receiver  |                |Telemetry Module
                  |                |
                  ▼                ▼
            +---------------------+
            | Power Distribution  |
            |    Board (PDB)      |
            +---------------------+
                     ▲
                     |
                  Battery
                     |
```

```
                    ▼
   ┌───────────┬───────────┬──────────┐
   ▼           ▼           ▼          ▼
 ESC 1       ESC 2       ESC 3      ESC 4
   │           │           │          │
   ▼           ▼           ▼          ▼
Motor 1     Motor 2     Motor 3    Motor 4
   │           │           │          │
   ▼           ▼           ▼          ▼
Propeller  Propeller  Propeller  Propeller

       All components are mounted on the
            Drone Frame (Chassis)
```

Components and Their Functions

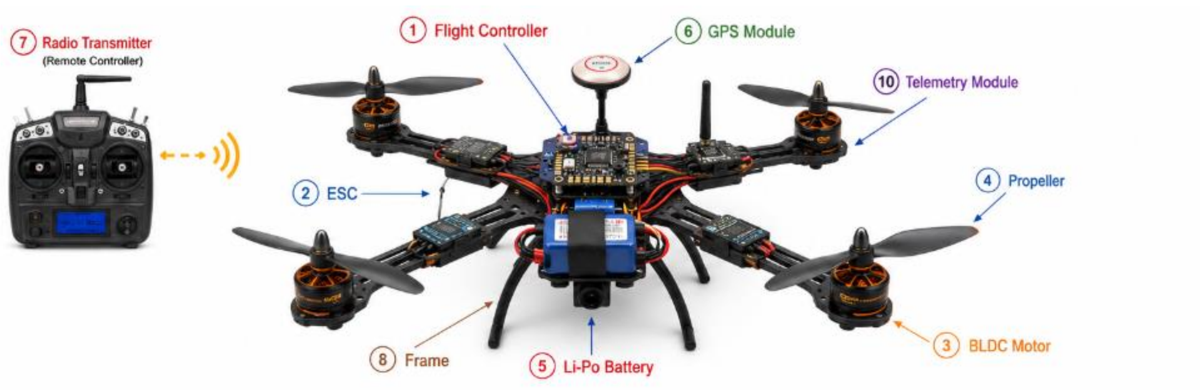
Component	Function
Flight Controller (FC)	The brain of the drone. It receives commands from the radio receiver and sensor data from the GPS and other sensors. It processes this information and sends control signals to the ESCs to stabilize and control the drone.
Electronic Speed Controller (ESC)	Controls the speed of each brushless motor by converting signals from the flight controller into the required electrical power. Each motor has its own ESC.
Brushless Motors	Convert electrical energy into mechanical rotation. They spin the propellers to generate lift and thrust. Brushless motors are efficient, reliable, and require low maintenance.
Propellers	Rotate to produce lift and thrust. By changing the speed of individual propellers, the drone can climb, descend, hover, move in different directions, and rotate (yaw).
Battery	Usually a Lithium Polymer (Li-Po) battery. It supplies electrical power to all electronic components, including the motors, flight controller, GPS, and telemetry module.
Power Distribution Board (PDB)	Distributes power from the battery to the ESCs, flight controller, GPS module, and other onboard electronics. It acts as the central power hub.
GPS Module	Provides the drone's geographical location, altitude, speed, and direction. It enables autonomous flight, waypoint navigation, Return-to-Home (RTH), and position hold.
Radio Receiver (RX)	Receives wireless control signals from the pilot's remote transmitter and forwards these commands to the flight controller.
Frame (Chassis)	The structural body of the drone. It supports and protects all components such as motors, battery, flight controller, GPS, and camera while maintaining balance and rigidity.
Telemetry Module	Enables wireless communication between the drone and the Ground Control Station (GCS). It transmits real-time flight data such as altitude, speed, battery voltage, GPS position, and system status.

Q7. Explain the role of the following components: flight controller, ESC, BLDC motor, propeller, Li-Po battery, GPS, radio transmitter, frame, battery eliminator circuit (BEC), telemetry module?

Answer7: The following are the major components of a drone and their functions:

Component	Role / Function
1. Flight Controller (FC)	The brain of the drone. It receives commands from the radio receiver and data from onboard sensors (GPS, gyroscope, accelerometer, etc.), processes them, and sends control signals to the ESCs to stabilize and control the drone during flight.
2. Electronic Speed Controller (ESC)	Controls the speed and direction of each BLDC motor. It converts the low-power control signals from the flight controller into high-current, three-phase power required by the motors. Each motor has its own ESC.
3. BLDC (Brushless DC) Motor	Converts electrical energy into mechanical rotation to spin the propellers. BLDC motors are efficient, lightweight, produce high torque, and require less maintenance than brushed motors.
4. Propeller	Generates lift and thrust when rotated by the BLDC motor. By varying the speed of individual propellers, the drone can take off, land, hover, move in different directions, and rotate (yaw).
5. Li-Po (Lithium Polymer) Battery	The main power source of the drone. It supplies electrical energy to the motors, flight controller, GPS, telemetry module, and other onboard electronics. Li-Po batteries are lightweight and have a high energy density, making them suitable for UAV applications.
6. GPS Module	Determines the drone's position, altitude, speed, and direction using satellite signals. It enables autonomous flight, waypoint navigation, position hold, geofencing, and Return-to-Home (RTH) functionality.
7. Radio Transmitter (Remote Controller)	Used by the pilot to send wireless flight commands such as throttle, pitch, roll, and yaw to the drone through the radio receiver. It allows manual control of the drone during flight.

Component	Role / Function
8. Frame (Chassis)	The structural body of the drone that supports and protects all components, including motors, battery, electronics, and payload. It provides strength, rigidity, and proper weight distribution while minimizing overall weight.
9. Battery Eliminator Circuit (BEC)	A voltage regulator that converts the high voltage from the Li-Po battery into a stable low voltage (typically 5 V or 12 V) required by components such as the flight controller, receiver, GPS, and telemetry module. It eliminates the need for separate batteries for these electronics.
10. Telemetry Module	Provides two-way wireless communication between the drone and the Ground Control Station (GCS). It transmits real-time flight information such as altitude, GPS coordinates, speed, battery voltage, and system status, and can also receive mission updates or configuration commands.



Q8. Explain the working principle and application of IMU, accelerometer, gyroscope, magnetometer, barometer, GPS, optical flow sensor, ultrasonic sensor, LiDAR, and camera. For each sensor mention purpose, working principle, and application?

Answer8:- Drone sensors provide information about the drone's position, orientation, altitude, speed, surroundings, and environment. The flight controller combines data from these sensors (sensor fusion) to achieve stable and autonomous flight.

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1. IMU (Inertial Measurement Unit)

Purpose:

- Measures the drone's motion and orientation.
- Maintains flight stability.

Working Principle:

- The IMU combines data from an accelerometer, gyroscope, and sometimes a magnetometer.
- It measures linear acceleration, angular velocity, and heading.

- The flight controller processes this data to estimate the drone's attitude (roll, pitch, and yaw).

Applications:

- Flight stabilization
- Attitude estimation
- Autonomous flight
- Navigation

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2. Accelerometer

Purpose:

- Measures linear acceleration along the X, Y, and Z axes.

Working Principle:

- Uses MEMS (Micro-Electro-Mechanical Systems) technology.
- When the sensor accelerates, a tiny proof mass moves, changing electrical capacitance.
- This change is converted into acceleration values.

Applications:

- Tilt detection
- Motion sensing
- Crash detection
- Flight stabilization

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3. Gyroscope

Purpose:

- Measures the drone's angular velocity.

Working Principle:

- Uses MEMS vibrating structures.
- The Coriolis effect measures rotational movement around the X, Y, and Z axes.

Applications:

- Roll, pitch, and yaw measurement
- Flight stabilization

- Camera gimbal stabilization

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4. Magnetometer

Purpose:

- Determines the drone's heading (electronic compass).

Working Principle:

- Detects the Earth's magnetic field using Hall-effect or magneto-resistive sensors.
- Calculates the direction of magnetic north.

Applications:

- Compass navigation
- Waypoint navigation
- Heading correction
- Return-to-Home (RTH)

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5. Barometer

Purpose:

- Measures altitude.

Working Principle:

- Measures atmospheric pressure.
- As altitude increases, air pressure decreases.
- The flight controller calculates altitude using pressure readings.

Applications:

- Altitude hold
- Smooth landing
- Vertical speed estimation

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6. GPS (Global Positioning System)

Purpose:

- Determines the drone's geographical location.

Working Principle:

- Receives signals from multiple satellites.
- Uses trilateration to calculate latitude, longitude, altitude, speed, and time.

Applications:

- Waypoint navigation
- Return-to-Home (RTH)
- Mapping
- Surveying
- Geotagging

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7. Optical Flow Sensor

Purpose:

- Measures movement relative to the ground.

Working Principle:

- A downward-facing camera captures continuous images.
- The sensor tracks movement of ground features between images to estimate speed and displacement.

Applications:

- Indoor navigation
- Position hold without GPS
- Precision landing
- Low-altitude flight

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8. Ultrasonic Sensor

Purpose:

- Measures the distance to nearby objects.

Working Principle:

- Emits ultrasonic sound waves.
- Measures the time taken for the echo to return.
- Distance is calculated using the time-of-flight principle.

Applications:

- Obstacle detection
- Altitude measurement
- Collision avoidance
- Indoor navigation

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9. LiDAR (Light Detection and Ranging)

Purpose:

- Measures distance accurately using laser light.

Working Principle:

- Emits laser pulses.
- Measures the time taken for the reflected light to return.
- Creates accurate 3D point clouds of the environment.

Applications:

- 3D mapping
- Terrain modeling
- Obstacle avoidance
- Autonomous navigation
- Forestry surveys

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10. Camera

Purpose:

- Captures images and videos.

Working Principle:

- A lens focuses light onto a CMOS or CCD image sensor.
- The sensor converts light into digital images or video.

Applications:

- Aerial photography
- Video recording
- Surveillance
- Object detection
- Photogrammetry

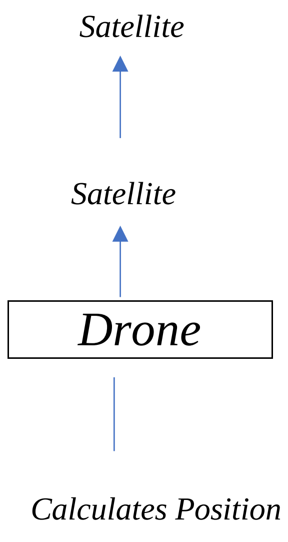
- Infrastructure inspection

Q9. Compare: (a) GPS vs Optical Flow, (b) LiDAR vs Ultrasonic, (c) Gyroscope vs Accelerometer?

ANSWER9:

GPS (Global Positioning System)	Optical Flow Sensor
• Uses signals from satellites to determine the drone's position.	• Uses a downward-facing camera to detect movement relative to the ground.
• Works best outdoors with a clear view of the sky.	• Works both indoors and outdoors at low altitude.
• Provides global position (latitude, longitude, altitude).	• Provides relative movement and velocity over the ground.
• Accuracy is typically 1–3 m (better with RTK GPS).	• Accuracy is high at low altitude (typically below 10 m).
• May lose accuracy near buildings or under dense tree cover.	• Performance decreases over smooth, reflective, or poorly textured surfaces.
• Used for waypoint navigation, Return-to-Home (RTH), mapping, and surveying.	• Used for indoor navigation, position hold, and precision landing.

GPS Navigation



Optical Flow

Drone

↓

Camera captures

ground images

↓

Tracks movement

between images

↓

Estimates motion

(b) LiDAR vs Ultrasonic Sensor

LiDAR (Light Detection and Ranging)	Ultrasonic Sensor
Uses laser light to measure distance.	Uses ultrasonic sound waves to measure distance.
Works using the time-of-flight of laser pulses.	Works using the echo time of sound waves.
Long measurement range (tens to hundreds of metres).	Short measurement range (typically 0.02–5 m).
Very high accuracy and resolution.	Moderate accuracy.
Can generate detailed 3D point clouds.	Measures only the distance to nearby objects.
More expensive.	Less expensive.
Used for 3D mapping, autonomous navigation, obstacle avoidance, and terrain modeling.	Used for obstacle detection, altitude measurement, and indoor navigation.

Simple Diagram

```
LiDAR

     Drone
       |
   Laser Pulse
       ▼
     Object
       ▲
 Reflected Light
       |
 Distance Calculated
Ultrasonic Sensor

     Drone
```

```
        |
 Sound Wave
        ▼
    Object
        ▲
 Echo Returns
        |
 Distance Calculated
```

(c) Gyroscope vs Accelerometer

Gyroscope	Accelerometer
Measures angular velocity (rotation).	Measures linear acceleration.
Detects rotation about the X, Y, and Z axes.	Detects motion along the X, Y, and Z axes.
Uses the Coriolis effect in MEMS sensors.	Uses a MEMS proof mass that moves during acceleration.
Helps maintain drone orientation.	Detects tilt, vibration, and movement.
Cannot determine position by itself.	Can estimate tilt using gravity when stationary.
Used for flight stabilization and gimbal control.	Used for motion sensing, attitude estimation, and crash detection.

Simple Diagram

```
Gyroscope

        ↺
     Rotation
        |
     [Drone]

Measures Roll, Pitch, Yaw
Accelerometer

          ↑ Z
          |
          |
Y ◄───────┼───────► X

Measures linear
acceleration along
X, Y, and Z axes.
```

Q10. Explain the concept of 6 Degrees of Freedom (6DOF). Include translation, rotation, X-axis, Y-axis, Z-axis. Draw suitable diagrams?

Answer10: The 6 Degrees of Freedom (6DOF) describe the six independent ways in which a drone (or any rigid body) can move in three-dimensional (3D) space. These include three translational (linear) movements and three rotational (angular) movements.

Understanding 6DOF is essential for controlling drone flight, navigation, and stabilization.

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1. Translation (Linear Motion)

Translation means the drone moves from one position to another without rotating.

(a) X-axis – Forward and Backward

```
         Forward (+X)
              ↑
              |
        ┌──────────┐
        │  Drone   │
        └──────────┘
              |
              ↓
        Backward (-X)
```

Explanation:

- +X: Drone moves forward.
- −X: Drone moves backward.

(b) Y-axis – Left and Right

```
Left (-Y)   ◄──────   ┌──────────┐   ──────►   Right (+Y)
                      │  Drone   │
                      └──────────┘
```

Explanation:

- +Y: Drone moves right.
- −Y: Drone moves left.

(c) Z-axis – Up and Down

```
          Up (+Z)
             ▲
             |
       ┌──────────┐
```

```
          │    Drone    │
          └─────────────┘
                 │
                 ▼
           Down (-Z)
```

Explanation:

- +Z: Drone ascends (takes off).
- −Z: Drone descends (lands).

---

2. Rotation (Angular Motion)

Rotation means the drone rotates about one of its three axes.

(a) Roll – Rotation about the X-axis

```
                 Front

          ┌─────────────┐
          │   Drone     │
          └─────────────┘

         ↺ Roll Left
         ↻ Roll Right

Rotation about X-axis
```

Explanation:

- Left side moves up or down.
- Right side moves in the opposite direction.
- Used for left/right banking during turns.

(b) Pitch – Rotation about the Y-axis

```
                Front
```

```
    ▲ Nose Up
      ↺
 ┌──────────────┐
 │ Drone        │
 └──────────────┘

      ↻
    ▼ Nose Down

Rotation about Y-axis
```

Explanation:

- Nose tilts upward or downward.
- Controls forward and backward movement.

(c) Yaw – Rotation about the Z-axis

```
              ↑

      ↺
  ┌──────────┐
  │  Drone   │
  └──────────┘
      ↻

Rotation about Z-axis
```

Explanation:

- Drone rotates left or right without changing its position.
- Used to change the heading (direction) of the drone.

Complete 6DOF Diagram

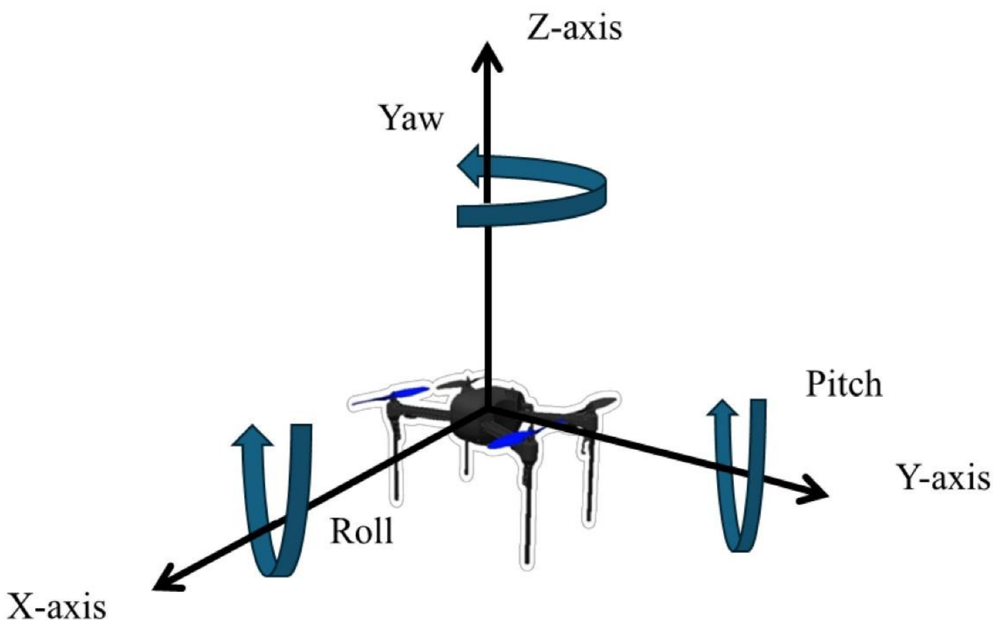
```
                   +Z  (Up)
                     ↑
                     │
        -Y ◄──────── Drone ────────► +Y
         (Left)                     (Right)

                     │
                     ▼
                  -Z  (Down)

            +X = Forward
            -X = Backward
```

```
Rotations

Roll  → Rotation about X-axis
Pitch → Rotation about Y-axis
Yaw   → Rotation about Z-axis
```



Q11. Explain roll, pitch, yaw, surge, sway, and heave with examples of each movement in a drone?

Answer11: A drone moves in six degrees of freedom (6DOF). These movements are divided into:

1. Rotational Movements (Angular Motion): Roll, Pitch, and Yaw
2. Translational Movements (Linear Motion): Surge, Sway, and Heave

These movements are controlled by varying the speed of the drone's motors through the flight controller.

---

1. Roll (Rotation about the X-axis)

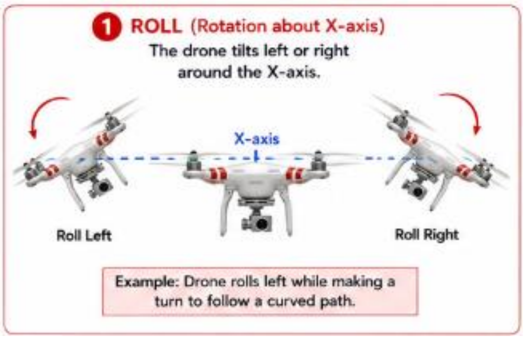
Definition:
Roll is the rotation of the drone about its X-axis, causing it to tilt to the left or right.

Diagram

```
            Front

         ↺ Roll Left
      ┌────────────┐
      │   Drone    │
      └────────────┘
         ↻ Roll Right

Rotation about X-axis
```



Example:

- A drone rolls left to move around an obstacle or rolls right while making a turn during aerial filming.

---

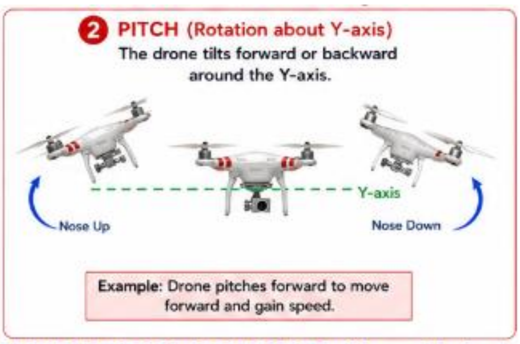
2. Pitch (Rotation about the Y-axis)

Definition:
Pitch is the rotation of the drone about its Y-axis, causing the nose to tilt upward or downward.

Diagram

```
               Nose Up
                  ▲
                  |
            ┌──────────┐
            │  Drone   │
            └──────────┘
                  |
                  ▼
              Nose Down

Rotation about Y-axis
```



Example:

- The drone pitches forward to accelerate and fly forward.
- It pitches backward to slow down or move backward.

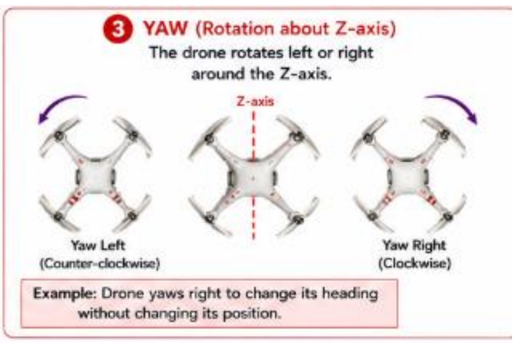
---

3. Yaw (Rotation about the Z-axis)

Definition:
Yaw is the rotation of the drone about its vertical (Z) axis without changing its position.

Diagram

```
           ↑
          ↺
     ┌──────────┐
     │  Drone   │
     └──────────┘
          ↻
Rotation about Z-axis
```



Example:

- While hovering, the drone rotates to face a building for inspection or changes its heading during aerial photography.

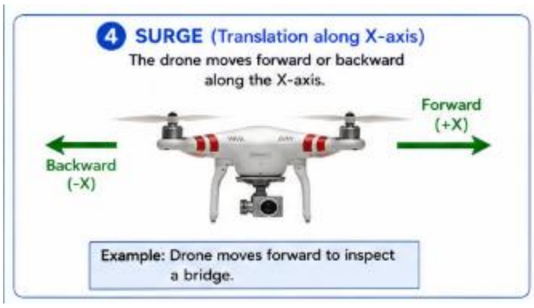
---

4. Surge (Translation along the X-axis)

Definition:
Surge is the forward or backward linear movement of the drone along the X-axis.

Diagram

```
 Forward (+X)
     ↑
  [ Drone ]
     ↓
Backward (-X)
```



Example:

- Flying forward to inspect a bridge.
- Flying backward to maintain a safe distance from an obstacle.

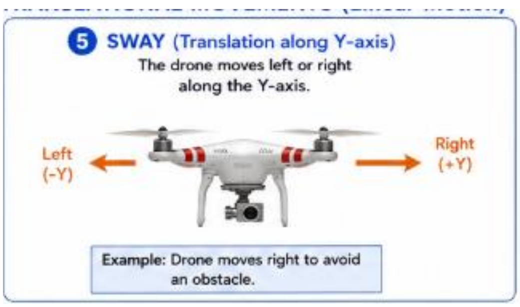
---

5. Sway (Translation along the Y-axis)

Definition:
Sway is the left or right linear movement of the drone along the Y-axis.

Diagram

```
Left (-Y) ←——     [ Drone ]     ——→ Right (+Y)
```



Example:

- Moving sideways to avoid a tree.
- Sliding right while recording a cinematic video.

---

6. Heave (Translation along the Z-axis)

Definition:
Heave is the vertical movement of the drone along the Z-axis.

Diagram

```
      Up (+Z)
        ▲
        |
   ┌──────────┐
   │  Drone   │
   └──────────┘
        |
        ▼
    Down (-Z)
```

Example:

- Ascending during take-off.
- Descending while landing.

Combined 6DOF Diagram

```
                 +Z (Heave Up)
                    ▲
                    |
                    |
-Y (Sway Left) ← Drone → +Y (Sway Right)
                    |
                    |
                    ▼
                 -Z (Heave Down)
```

```
          +X = Surge Forward
          -X = Surge Backward

Rotations:
• Roll  → Rotation about X-axis
• Pitch → Rotation about Y-axis
• Yaw   → Rotation about Z-axis
```

Q12. Explain the complete signal flow in a drone: Pilot Input to Radio Transmitter to Receiver to Flight Controller to ESC to BLDC Motors to Propellers to Drone Movement. Illustrate with a block diagram?

Answer12: The signal flow in a drone describes how the pilot's commands travel from the remote controller to the drone's motors, resulting in controlled flight. The flight controller acts as the brain of the drone, processing commands and sensor data before controlling the motors.

Block Diagram of Complete Signal Flow

```
        Pilot
          ↓
 [Radio Transmitter]
          |
      RF Signals
          ↓
 [Radio Receiver]
          ↓
 [Flight Controller] ←──────────┐
          ↓                     │
 [ESC (Electronic               │
  Speed Controller)]            │
          ↓                     │
 [BLDC Motors]                  │
          ↓                     │
 [Propellers]                   │
          ↓                     │
 [Drone Movement]               │
                                ┘
```

```
                                  |
       IMU • GPS • Barometer • Magnetometer
         (Sensor Feedback to Flight Controller)
```

Signal Flow with Pictures


GEPRC
915M 500mW
R T
5V G
LED

4

Step-by-Step Explanation

1. Pilot Input

- The pilot moves the joysticks on the remote controller.
- Commands include:
  - Throttle (up/down)
  - Pitch (forward/backward)
  - Roll (left/right)
  - Yaw (rotation)

Function: Generates the desired flight command.

---

2. Radio Transmitter (Remote Controller)

- Converts joystick movements into Radio Frequency (RF) signals.
- Transmits these signals wirelessly to the drone.

Function: Sends pilot commands to the drone.

---

3. Radio Receiver

- Mounted on the drone.
- Receives RF signals from the transmitter.
- Converts them into digital control signals.

Function: Passes pilot commands to the flight controller.

---

4. Flight Controller (Brain of the Drone)

- Receives:
  - Pilot commands
  - Sensor data (IMU, GPS, Barometer, Magnetometer)
- Calculates the required motor speeds.
- Sends commands to the ESCs.

Function: Controls and stabilizes the drone.

---

5. Electronic Speed Controller (ESC)

- Receives control signals from the flight controller.
- Converts battery power into controlled three-phase electrical power for each BLDC motor.
- Adjusts motor speed according to the command.

Function: Controls the speed of each motor.

---

6. BLDC (Brushless DC) Motors

- Receive power from the ESCs.
- Convert electrical energy into rotational motion.

Function: Rotate the propellers.

---

7. Propellers

- Spin at different speeds.
- Produce lift and thrust.

Function: Generate the forces required for flight.

---

8. Drone Movement

By varying the speed of individual motors, the drone performs:

- Take-off
- Landing
- Hovering
- Forward flight
- Backward flight
- Left and right movement
- Roll, Pitch, and Yaw

---

9. Sensor Feedback (Closed-Loop Control)

Sensors continuously send data back to the flight controller:

- IMU: Measures acceleration and rotation.
- GPS: Provides position and navigation.
- Barometer: Measures altitude.
- Magnetometer: Provides heading (compass).

The flight controller compares the pilot's commands with the sensor data and continuously adjusts motor speeds to maintain stable flight.

Complete Signal Flow Sequence

```
Pilot
   │
   ▼
Radio Transmitter
   │
   ▼
Radio Receiver
   │
   ▼
Flight Controller
   │
   ▼
ESC
   │
   ▼
BLDC Motors
   │
   ▼
Propellers
```

```
    |
    ▼
Drone Movement
    ▲
    |
Sensor Feedback (IMU, GPS, Barometer, Magnetometer)
```

Q13. Why is PID control important in drones? Explain the proportional, integral, and derivative terms?

Answer13: PID (Proportional–Integral–Derivative) control is one of the most important control algorithms used in drones. It enables the flight controller to automatically maintain the drone's stability, balance, altitude, and direction by continuously correcting errors between the desired position and the actual position.

Without PID control, a drone would become unstable, making it difficult or impossible to hover or fly accurately.

Why is PID Control Important?

PID control helps the drone to:

- Maintain stable flight.
- Hover at a fixed position.
- Correct disturbances caused by wind.
- Achieve smooth take-off and landing.
- Improve flight accuracy.
- Reduce oscillations and overshooting.
- Enable autonomous navigation and waypoint missions.

Example:
If the pilot commands the drone to hover at 10 m, but the drone rises to 11 m, the PID controller automatically adjusts the motor speeds to bring the drone back to 10 m.

Working Principle of PID Control

The PID controller continuously performs the following steps:

1. Measures the current position using sensors (IMU, GPS, Barometer).
2. Compares it with the desired position.
3. Calculates the error.
4. Computes the correction using the PID algorithm.
5. Adjusts the speed of the motors through the ESCs.

Block Diagram of PID Control

```
          Desired Position
                 |
                 ▼
        ┌─────────────────┐
        │ Compare Error   │
        └─────────────────┘
                 |
                 ▼
          PID Controller
      (P + I + D Algorithm)
                 |
                 ▼
        Flight Controller
                 |
                 ▼
               ESCs
                 |
                 ▼
           BLDC Motors
                 |
                 ▼
            Drone Motion
                 ▲
                 |
        Sensor Feedback
(IMU, GPS, Barometer, etc.)
```

1. Proportional (P) Control

Definition

The Proportional (P) term produces an output that is directly proportional to the current error.

Formula

$$P = K_p \times Error$$

where:

- Kp = Proportional gain
- Error = Desired Position − Actual Position

Working

- Large error → Large correction
- Small error → Small correction

Example

Desired altitude = 10 m

Actual altitude = 8 m

Error = 2 m

The proportional controller immediately increases motor speed to raise the drone.

Advantages

- Quick response
- Easy implementation

Limitation

High proportional gain can cause overshoot and oscillations.

---

2. Integral (I) Control

Definition

The Integral (I) term accumulates the error over time and removes any steady-state error.

Formula

$$I = K_i \int Error\, dt$$

where:

- Ki = Integral gain

Working

If the drone remains slightly below the desired altitude for several seconds, the integral term gradually increases the correction until the error becomes zero.

Example

Desired altitude = 10 m

Drone continuously flies at 9.8 m

The integral controller slowly increases motor thrust until the drone reaches exactly 10 m.

Advantages

- Eliminates steady-state error
- Improves accuracy

Limitation

Too much integral action may cause oscillation and integral windup.

---

3. Derivative (D) Control

Definition

The Derivative (D) term predicts future error by measuring the rate of change of the error.

Formula

$$D = K_d \times \frac{d(Error)}{dt}$$

where:

- Kd = Derivative gain

Working

If the drone is approaching the desired altitude too quickly, the derivative controller slows the motors before overshooting.

Example

The drone is climbing rapidly toward 10 m.

The derivative controller reduces motor speed to prevent the drone from rising above the target altitude.

Advantages

- Reduces overshoot
- Improves stability
- Dampens oscillations

Limitation

Sensitive to sensor noise.

Combined PID Equation

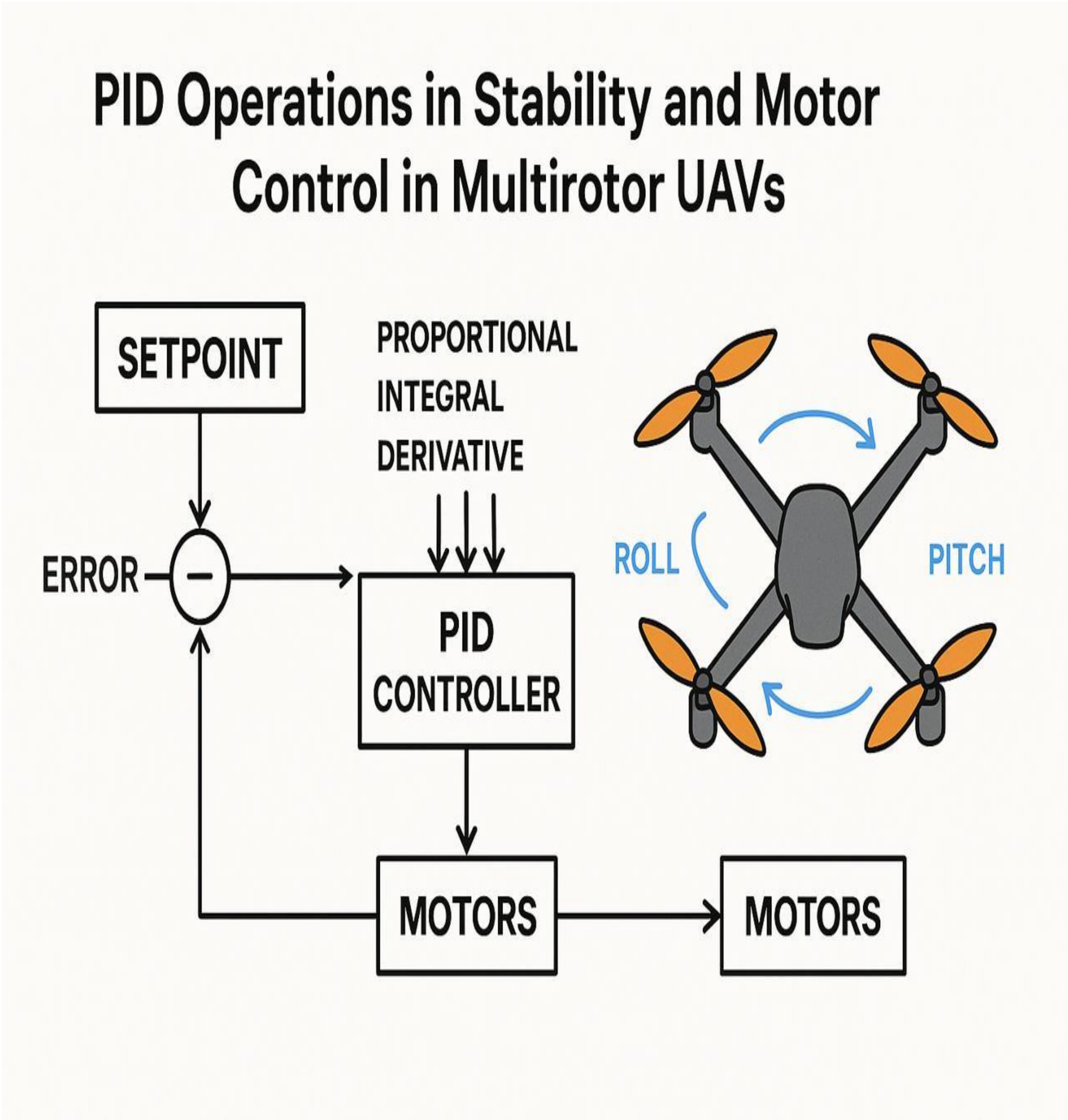
The overall PID controller output is:

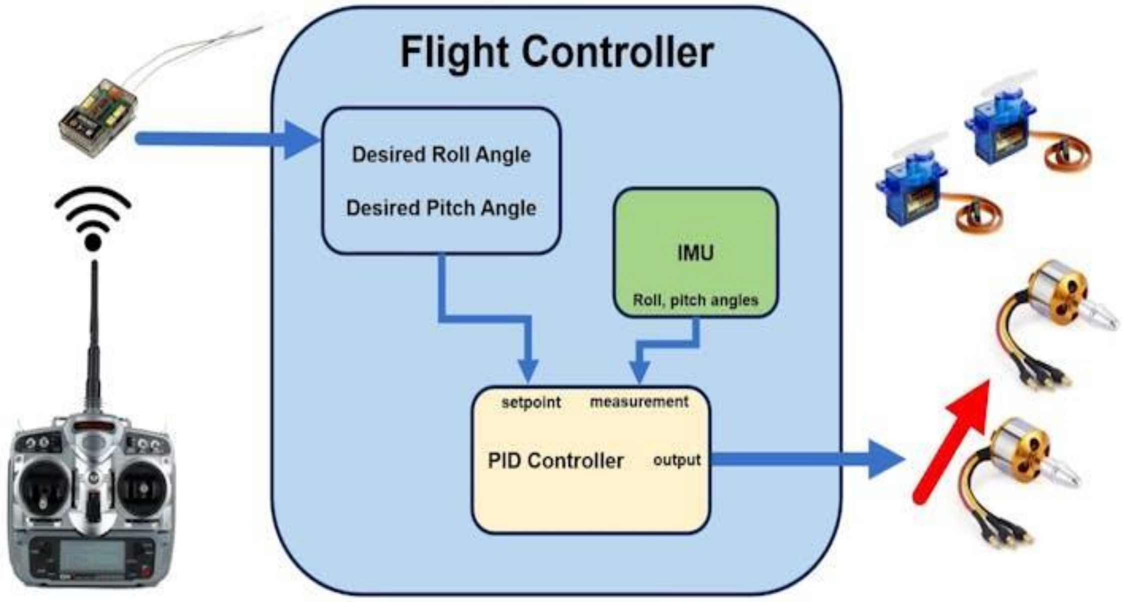
$$\boxed{\text{Output} = K_p \times Error + K_i \int Error\, dt + K_d \frac{d(Error)}{dt}}$$

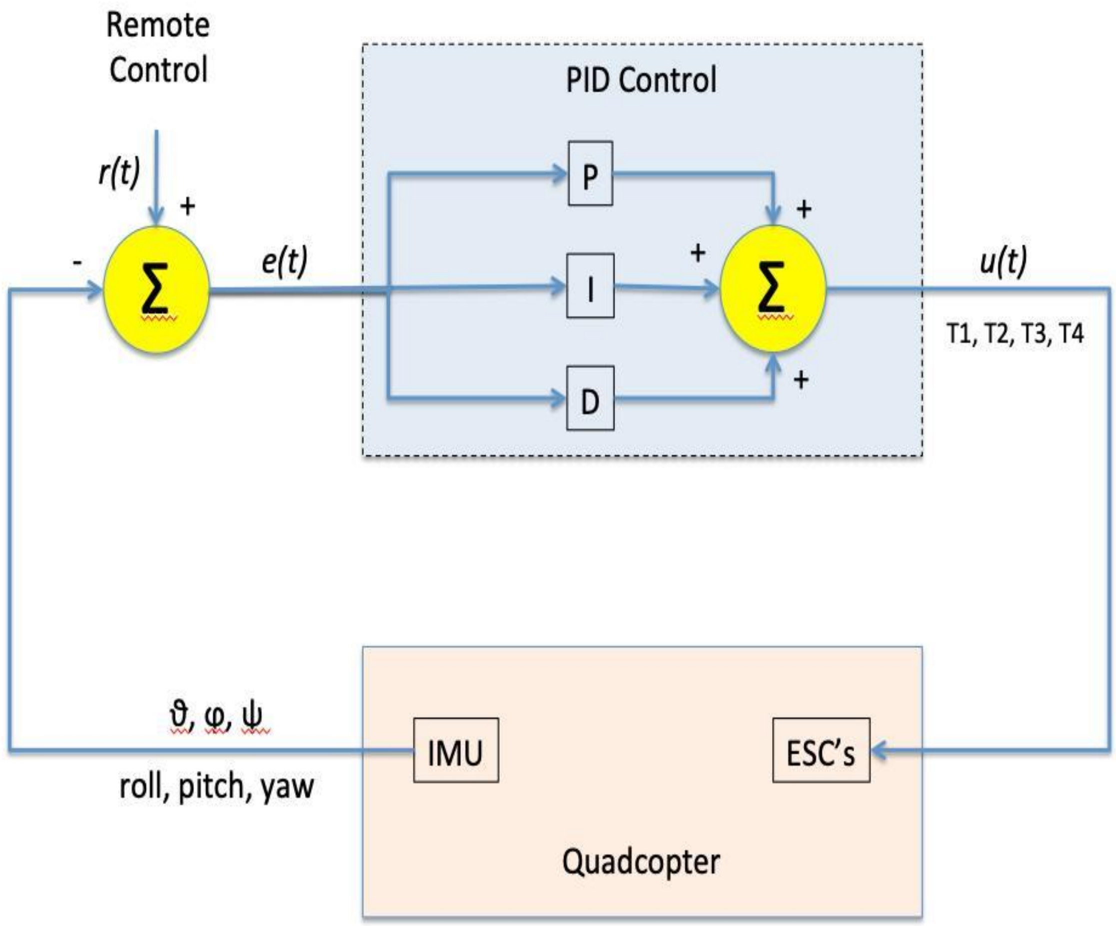
PID Controller Illustration

```
                Error
                  |
      ------------+------------
      |           |           |
      v           v           v
Proportional   Integral   Derivative
 (Present)      (Past)     (Future)
      |           |           |
      ------------+------------
                  v
            PID Controller
                  |
                  v
         Motor Speed Control
                  |
                  v
            Stable Flight
```

Illustration of PID Control in a Drone


PID Operations in Stability and Motor Control in Multirotor UAVs
SETPOINT
PROPORTIONAL
INTEGRAL
DERIVATIVE
ERROR
−
PID
CONTROLLER
ROLL
PITCH
MOTORS
MOTORS

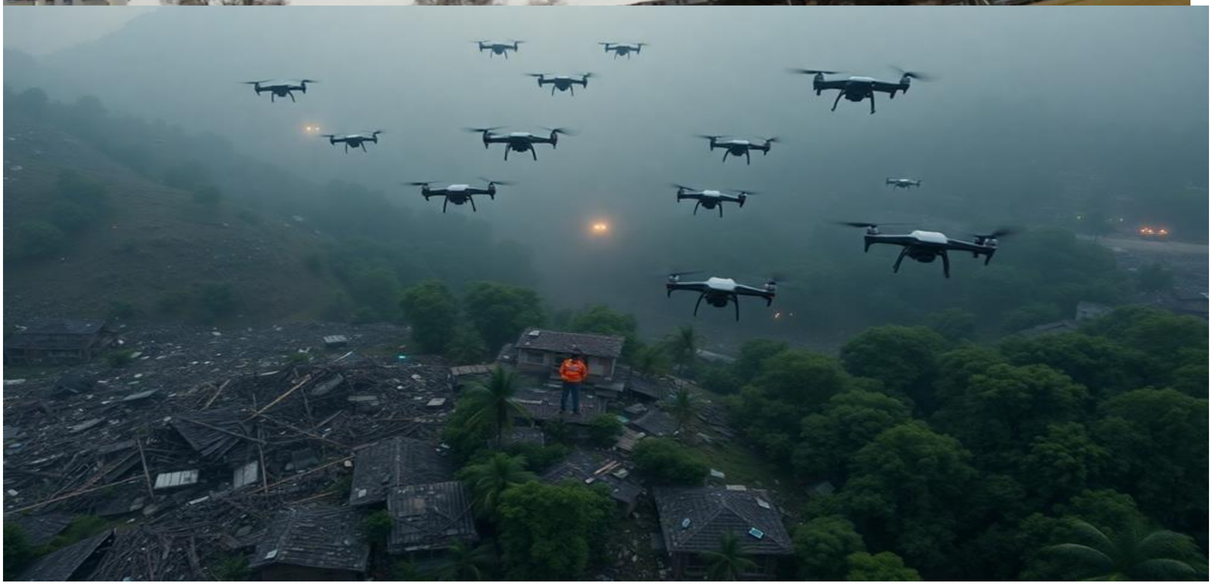

Flight Controller
Desired Roll Angle
Desired Pitch Angle
IMU
Roll, pitch angles
setpoint
measurement
PID Controller
output

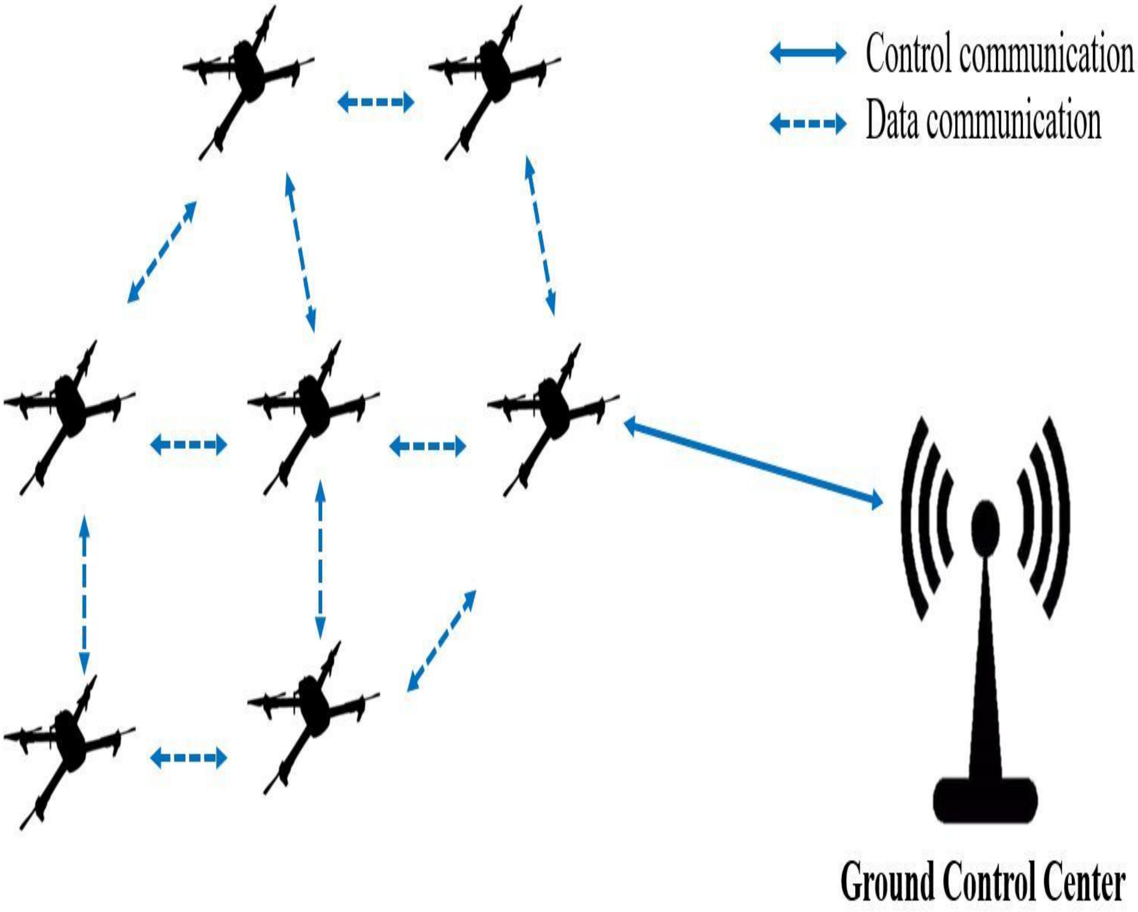

Remote Control
PID Control
r(t)
+
-
Σ
e(t)
P
I
D
+
+
+
Σ
u(t)
T1, T2, T3, T4
ϑ, φ, ψ
roll, pitch, yaw
IMU
ESC's
Quadcopter

Q14. Bonus — Research Questions. Answer any of the following:

- Explain drone swarm technology.
- What is autonomous navigation?

Answer14: 1. Drone Swarm Technology





6

Drone swarm technology is the coordination of multiple drones that work together as a single intelligent system. Instead of each drone operating independently, the swarm communicates, shares information, and cooperates to complete tasks efficiently.

How it works

A drone swarm generally consists of four main components:

1. Communication
   - Drones exchange information through wireless communication (e.g., Wi-Fi, radio, or mesh networks).
   - They share position, speed, battery status, and mission updates.
2. Coordination
   - Algorithms assign tasks and organize formations.
   - The swarm maintains safe spacing to avoid collisions.
3. Decision-Making
   - The swarm may use centralized control (one controller directs all drones) or decentralized control (each drone makes local decisions based on shared information).
4. Execution
   - Drones cooperate to complete the mission, adapting if one drone fails or conditions change.

Applications

- Search and rescue operations
- Disaster assessment
- Precision agriculture
- Environmental monitoring
- Military reconnaissance
- Warehouse inventory
- Entertainment (drone light shows)
- Large-area mapping and surveying

Advantages

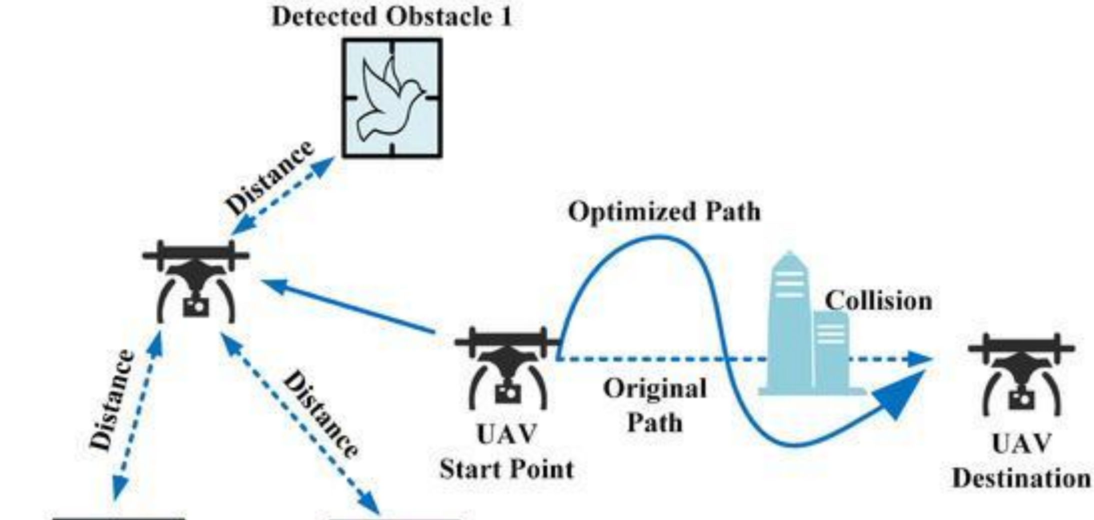
- Covers larger areas faster.
- Increases reliability through redundancy.
- Can continue operating even if individual drones fail.
- Reduces mission time.
- Scales to larger operations by adding more drones.

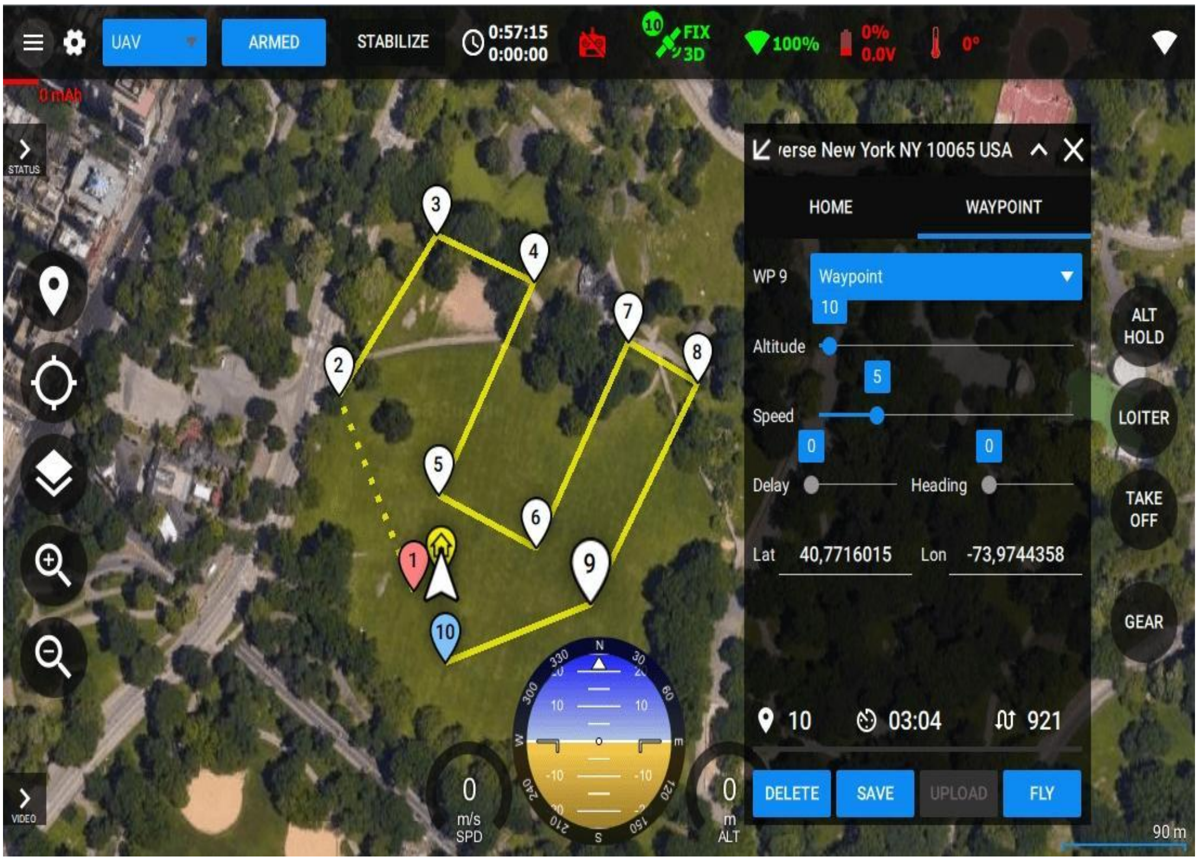
Challenges

- Reliable communication between drones.
- Collision avoidance within the swarm.
- Synchronization and task allocation.
- Battery limitations.
- Cybersecurity and protection against interference.

---

2. Autonomous Navigation


Detected Obstacle 1
Distance
Optimized Path
Collision
Distance
Distance
Original
Path
UAV
Start Point
UAV
Destination
Detected Obstacle 3
Detected Obstacle 2



Autonomous navigation is the ability of a drone to fly and complete a mission without continuous human control. The drone determines its position, plans a route, avoids obstacles, and reaches its destination automatically.

How autonomous navigation works

The navigation process typically includes:

1. Localization
    - Determines the drone's current position using technologies such as:
        - GPS/GNSS
        - Inertial Measurement Unit (IMU)
        - Compass (magnetometer)
        - Barometer
        - Visual odometry
        - SLAM (Simultaneous Localization and Mapping)
2. Environment Perception
    - Detects obstacles and terrain using:
        - Cameras
        - LiDAR
        - Radar
        - Ultrasonic sensors
        - Infrared sensors
3. Path Planning
    - Calculates an efficient and safe route.

    - Can dynamically re-plan if new obstacles appear.
4. Flight Control
    - The flight controller adjusts the motors to follow the planned path while maintaining stability.

Features

- Automatic takeoff and landing
- Waypoint navigation
- Obstacle avoidance
- Return-to-home (RTH)
- Terrain following
- Dynamic route replanning
- Precision hovering

Applications

- Package delivery
- Infrastructure inspection
- Precision agriculture
- Search and rescue
- Surveillance
- Mapping and surveying
- Wildlife monitoring

Advantages

- Reduces pilot workload.
- Improves safety by minimizing human error.
- Enables operation in hazardous or remote areas.
- Produces consistent, repeatable flight paths.
- Supports beyond-visual-line-of-sight (BVLOS) missions where regulations permit.

Limitations

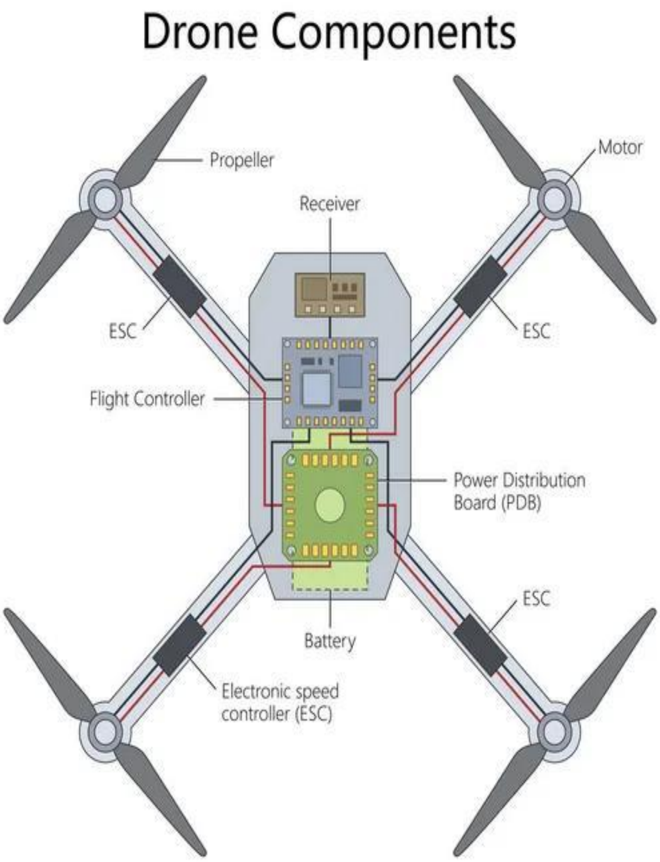
- GPS signals may be weak or unavailable in urban canyons or indoors.
- Adverse weather can reduce sensor performance.
- Dynamic environments require fast perception and planning.
- Autonomous systems depend on reliable sensors, software, and communications.

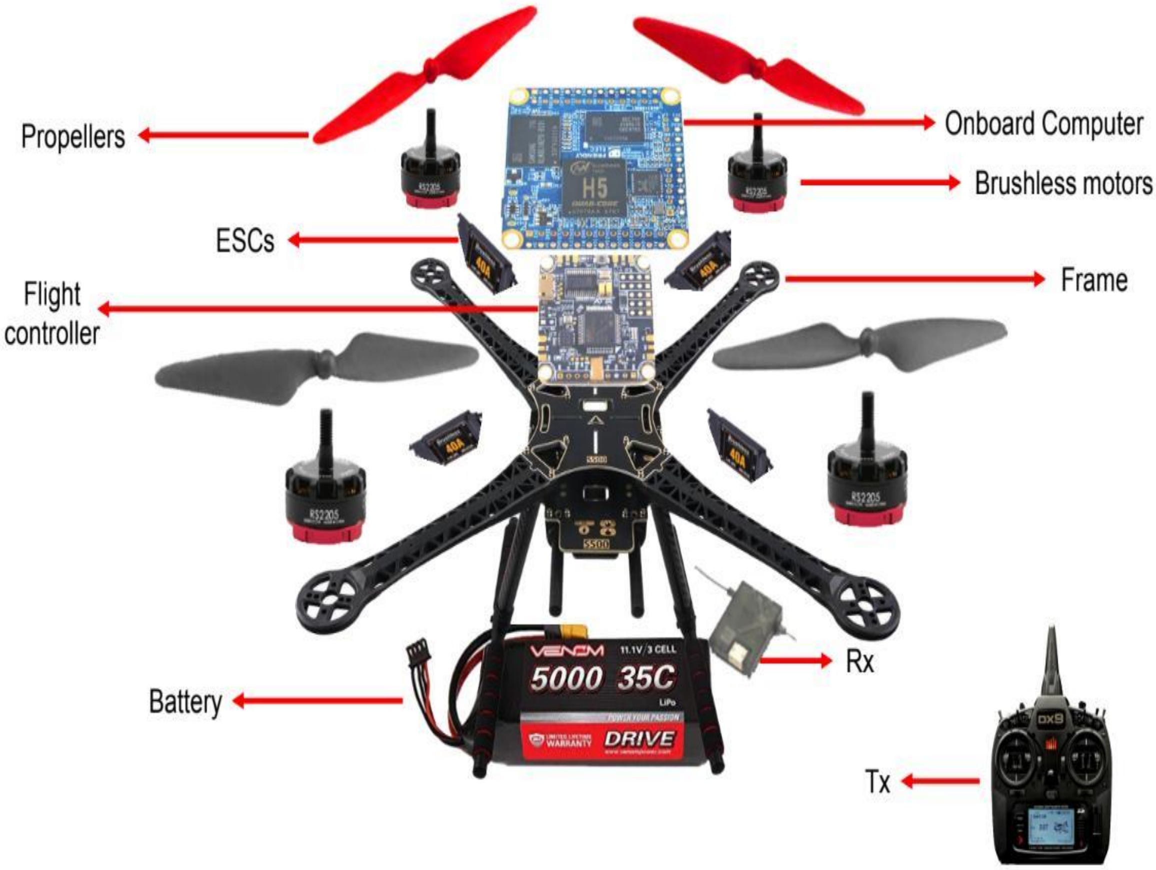
15. Practical Activity —

Identify and label every component of a real drone or a drone image.

Answer:15


PROPELLERS
FRAME
ELECTRONIC SPEED CONTROLLER
MOTORS
POWER DISTRIBUTION BOARD
FLIGHT CONTROLLER
BATTERY
CAMERA


Drone Components
Propeller
Motor
Receiver
ESC
ESC
Flight Controller
Power Distribution Board (PDB)
ESC
Battery
Electronic speed controller (ESC)



5

Objective

Identify and label the main components of a quadcopter drone and understand the function of each part.

Components and Their Functions

Component	Function
Frame	The main structure that supports and connects all drone components.
Propellers	Generate lift and thrust, enabling the drone to fly.
Brushless Motors	Rotate the propellers at varying speeds to control movement.
Electronic Speed Controllers (ESCs)	Regulate the speed of each motor based on commands from the flight controller.
Flight Controller (FC)	The "brain" of the drone that processes sensor data and controls flight stability.
Battery (Li-Po)	Supplies electrical power to the drone.
Power Distribution Board (PDB)	Distributes power from the battery to the ESCs and other electronics (may be integrated into modern flight controllers).
GPS Module	Provides positioning, navigation, waypoint flight, and Return-to-Home (RTH) functionality.

Component	Function
Receiver (RX)	Receives control signals from the remote controller.
Transmitter (Remote Controller)	Sends flight commands to the drone (external to the drone).
Camera	Captures photos and videos or provides live video for FPV.
Gimbal	Stabilizes the camera to produce smooth images and videos.
Antennas	Enable radio communication, GPS reception, and video transmission.
IMU (Inertial Measurement Unit)	Measures acceleration and angular motion to help stabilize the drone.
Compass (Magnetometer)	Determines the drone's heading relative to Earth's magnetic field.
Obstacle Avoidance Sensors	Detect nearby objects to help prevent collisions.
Landing Gear	Supports the drone during take off and landing while protecting the camera and frame.

Q:Higher-Order Thinking Questions

- *Reflect on and answer the following in your own words:*

  Q1:- Why do quadcopters require four motors instead of one?

Answer1: A quadcopter needs four motors to fly, stay balanced, and move in different directions.

- One motor can only make the drone go up, but it cannot keep it balanced.
- Four motors work together to keep the drone stable and allow it to move safely.

Each motor helps control the drone:

- All four motors spin faster → The drone goes up.
- All four motors spin slower → The drone goes down.
- Front motors slower, back motors faster → The drone moves forward.
- Left motors faster, right motors slower → The drone moves right (and vice versa).
- Some motors spin faster than others → The drone turns left or right.

Q2:What should happen if one motor stops working during flight?

Answer2: If one motor stops working, the drone becomes unbalanced and may lose control.

- The drone may tilt or spin.
- It may fall to the ground if it cannot stay balanced.
- Some advanced drones can detect the problem and land safely or return home if possible.

Q3: Why are brushless motors preferred over brushed motors?

Answer3: Brushless motors are preferred because they are more powerful, efficient, and last longer than brushed motors.

Advantages of Brushless Motors

- More efficient – Use less battery power.
- More powerful – Produce higher speed and lift.
- Longer life – No brushes to wear out.
- Less maintenance – Require little servicing.
- Quieter operation – Make less noise.

Disadvantages of Brushed Motors

- Brushes wear out over time.
- Less efficient and waste more energy.
- Produce more heat.
- Have a shorter lifespan.

Q4:Can a drone fly indoors without GPS ?explain.

Answer4: Yes, a drone can fly indoors without GPS.

GPS signals are usually weak or unavailable indoors, so the drone cannot use GPS for positioning.

Instead, the drone uses:

- Sensors (such as cameras or optical flow sensors)
- Gyroscope and accelerometer (IMU) to maintain balance
- Ultrasonic or LiDAR sensors to measure height and avoid obstacles

However, if the drone does not have these sensors, it may drift and require careful manual control.

Q5:Why are Li-Po batteries used in drones instead of standard batteries?

Answer5: Li-Po batteries are used in drones because they are lightweight, provide high power, and help the drone fly longer and better than standard batteries.

Q6: What factors reduce a drone's flight time?

Answer6**:** A drone's flight time is reduced by heavy loads, strong winds, high speed, an old battery, cold weather, and frequent rapid movements because these make the drone use more battery power**.**

Q7: How can drone stability be improved during strong winds?

Answer7: Drone stability in strong winds can be improved by flying lower, flying smoothly, using GPS and sensors, keeping the battery charged, and avoiding very windy conditions.

Q8: Why is sensor fusion important in drones?

Answer8: Sensor fusion means combining information from different sensors to help the drone fly more accurately and safely.

For example, a drone combines data from:

- GPS – Finds its location.
- IMU (gyroscope and accelerometer) – Keeps the drone balanced.
- Compass – Shows the direction.
- Barometer – Measures altitude.

By using all these sensors together, the drone gets a more accurate picture of its position and movement.

Benefits of Sensor Fusion

- Improves flight stability.
- Increases navigation accuracy.
- Helps avoid obstacles.
- Makes autonomous flight safer.
- Continues flying accurately even if one sensor is less reliable.

Q9: Explain how GPS and IMU work together for stable navigation?

Answer9: GPS and IMU work together to help a drone know where it is and stay balanced while flying.

- GPS (Global Positioning System) tells the drone its location and helps it follow a route.
- IMU (Inertial Measurement Unit) contains a gyroscope and accelerometer, which help the drone keep its balance and detect movement.

How they work together

1. GPS tells the drone where it is.
2. IMU tells the drone if it is tilting, turning, or accelerating.
3. The flight controller combines information from both sensors.
4. It adjusts the motor speeds to keep the drone stable and on the correct path.

Example

If a gust of wind pushes the drone:

- GPS detects that the drone has moved away from its position.
- IMU detects the tilt and movement immediately.
- The flight controller increases or decreases the speed of the motors to bring the drone back to its correct position.

Q10: If you were designing a drone for mountain rescue operations, which sensors and components would you choose, and why?

Answer10:For a mountain rescue drone ,I would choose the following sensors and components:

Component/Sensor	Why it is Needed
GPS	To find the drone's location and navigate to the rescue area.
IMU (Gyroscope & Accelerometer)	To keep the drone stable in strong mountain winds.
LiDAR or Obstacle Avoidance Sensors	To detect rocks, trees, and cliffs and avoid collisions.
High-Resolution Camera	To search for missing or injured people.
Thermal Camera	To detect body heat, especially at night or in fog.
Brushless Motors	To provide strong, reliable flight in windy conditions.
Li-Po Battery	To supply lightweight, high power for longer flights.
Radio Communication System	To send live video and location information to rescue teams.
LED Lights	To improve visibility during night operations.

These components help the drone:

- Fly safely in difficult mountain terrain.
- Avoid obstacles such as trees and cliffs.
- Find people using normal and thermal cameras.
- Stay stable in strong winds.
- Send live information to rescue teams.